Self-Study Assignment Data-Link CSE1405

Connor Spruit

Delf University of Technology, The Netherlands

1. **Question 1**

   The first one has 5 ones. The second has 10. The third has 5. the forth has 4. therefore rows 1 and 3 are guaranteed to be wrong.

2. **Question 2**

   No, there is no way to trace back which bit should be flipped. It is also not possible with 1 bit to detect if 2 bit flips have occured.

3. **Question 3**

   No, as discussed in question 2, it is possible 2 bits have been flipped.

4. **Question 4 (parity words)**

   No the reciever does not find any error. The reciever does not know if the sequence is correct as we are unable to tell if multiple bit flips occured.

5. **Question 5 (internet checksum)**

   add them up, add overflow back to the front (how the fuck). then flip bits add to the end.

6. **Question 6 (crc computation)**

   the division is 110 100 011 000 over 1101 which gives 1 0000 0000 & r=011. Therefore the message is 110 100 011 011.

7. **Question 7 (crc error detection)**

   1100 1110 0111 1000 01 over 110010 is 1000010000100 with remainder 11001. Thus there has been an error.

8. **Question 8 (crc - parity)**

   k = 5 - 1 = 4.
   Thuis 4 0's are added to the message befored deviding by the generator.
   this leads to 10001111 r=101.
   thus the message becomes 101101000101.
   this is even parity so a 0 is added to the end to create
   1011010001010.

9. Question 9 (parity blocks)

1 0 1 1 | 1 0 0 0 1 | 1 1 0 0 0 | 1 0 1 1 1 | 1 ----------- 0 1 0 1 | 0